# IMHIS

Impact Monitor for Health Information Systems

## Digital systems – real impact:

## What hospital staff truly needs

Dr. med. Florian Eisold, B.Sc., LL.M.

## IMHIS® at a glance

**What if the digital hospital of the future is meant to be a reality tomorrow? Then we would need to know today, what is future-ready and what is not.**

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# IMHIS® at a glance

## What

Analysis tool for **health information systems:** from paper lists to modular systems up to complex digital solutions such as hospital information systems

## Why

Assesses information systems from the **user perspective:** impact, usability, acceptance

## How

Analysis across **6** clearly defined **dimensions** – via **Questionnaires** and **Observation**: at the levels of people, system, organization

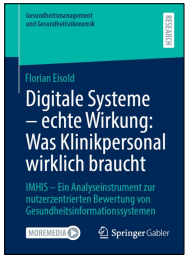
## Where

**Facility-independent applicability:** from medical practices, to rehabilitation facilities up to hospitals

## Value

**Effective digitalization** with **real added value: identify** weaknesses, **build** on strengths, **secure** digital impact

- ✓ **makes digital impact visible**
- ✓ **creates evidence that powers tomorrow's decisions**
- ✓ **closes the gulf between ambition and reality**




**IMHIS® – making digitalization work.**

**The book on the method:**

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# The promise of digitalization – and the reality

**Information quality**

***Expectation:*** current, complete and reliable data **available anytime**
***Reality:*** **scattered**, inconsistent and outdated data

**System quality**

***Expectation:*** stable, fast and **mobile-usable systems**
***Reality:*** **time and data losses** due to loading times, crashes and updates

**User satisfaction**

***Expectation:*** intuitive, automated and **workload-reducing systems**
***Reality:*** cluttered interfaces, poor usability, **frustration** in daily work

**Communication & Collaboration**

***Expectation:*** transparent and **digital collaboration** in real time
***Reality:*** digital **one-way** entries simulate collaboration without creating it

**Workload**

***Expectation:*** **less bureaucracy** and more time for patients
***Reality:*** **duplicate entries,** clickwork and missing intelligent workflows

**Work processes**

***Expectation:*** **intelligent**, automated and efficient processes
***Reality:*** **parallel worlds** of paper lists and non-integrated systems

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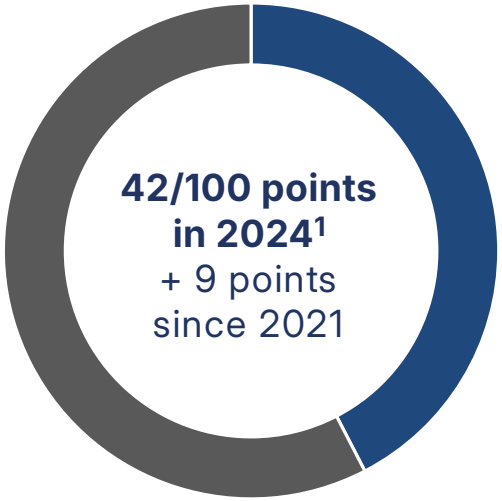
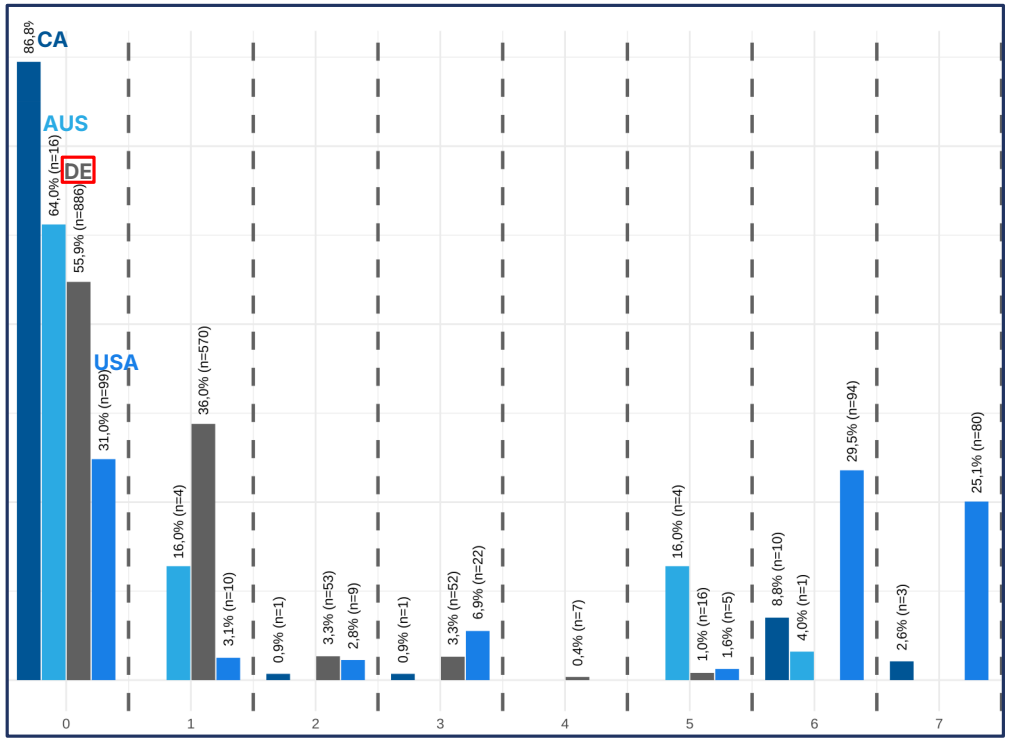
# Germany's healthcare system is lagging behind digitally

DigitalRadar




**58% unused digitalization potential**




**EMRAM-Scores in 2024:[1]**
Stufe 0: **no** Labor/Radiologie/Apotheken HIS
Level 7: "paperless" hospital

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## Germany’s healthcare system is lagging behind digitally

# Enough empty promises.
# The time for digital impact is now.

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# The crucial question

**Not:**

***"How digital are we?"***

**But:**

***"Does our digitalization deliver measurable impact?"***

**Reality**

**IMHIS**

**Expectation**

**Makes digital impact visible**

– and closes the gap between expectation and reality.

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# The solution: IMHIS®

## 6 dimensions that truly matter for digital impact

**Information quality**
- ✓ Timeliness
- ✓ Completeness
- ✓ Relevance

**Communication & Collaboration**
- Teamwork ✓
- Accessibility ✓
- Value ✓

**System quality**
- ✓ Stability
- ✓ Performance
- ✓ Functionality

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**Free | Evidence-based | Practical**

**Workload**
- Mental ✓
- Physical ✓
- Time ✓

**User satisfaction**
- ✓ Interface
- ✓ Usefulness
- ✓ Acceptance

**Work processes**
- What, when, where and how ✓
- Task duration ✓
- Workflow changes ✓

**IMHIS® combines 5 subjective questionnaires (99 items) with an objective observation (3 dimensions and 26 categories)**

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## The solution: IMHIS®

# Every software promises digital impact. IMHIS® proves it.

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# The digital patient chart Meona in the IMHIS® test

## The digital patient chart Meona is supposed to reduce media breaks, speed up processes and relieve staff – but does it work in practice?

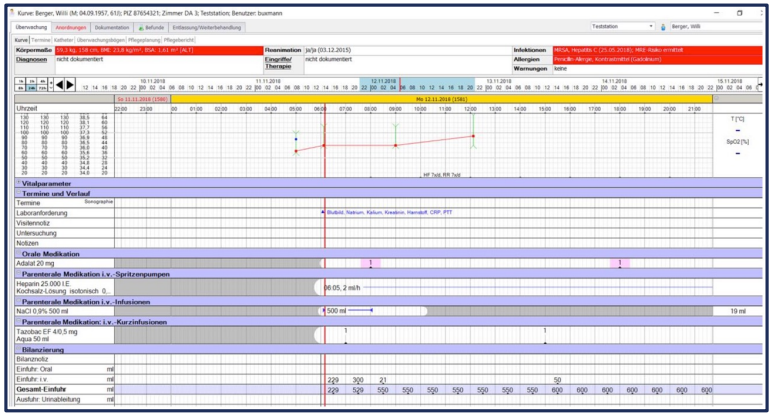
### Reality[2]

X **Scattered information:** multiple paper notes, limited readability and reliability

X **No shared use** → Delayed orders & information flow

X **No interfaces**, duplication of work and media breaks

### Expectation[3]



✓ **Everything in one place:** e.g. vitals, fluid management, pain, ward round documentation

✓ **Electronic medication** and physician **orders**

✓ **HIS integration:** lab/imaging and prior findings

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# The digital patient chart Meona in the IMHIS® test

**Digitalization of the patient chart**

*6 months apart*

**Pretest**
before implementation

**22 participants**
(= 68% response rate)

**120 hours**
(= 14 working days)

**General surgery ward**
(University hospital)

IMHIS

**5 questionnaires with 99 items**
- ✓ Information quality
- ✓ System quality
- ✓ User satisfaction
- ✓ Communication & collaboration
- ✓ Workload

**Observation with 26 categories**
- ✓ What, when, where and how
- ✓ Task duration
- ✓ Workflow changes

**Posttest**
after implementation

**23 participants**
(= 78% response rate)

**130 hours**
(= 15 working days)

**General surgery ward**
(University hospital)

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## The digital patient chart Meona in the IMHIS® test

**What we need is a digitalization that brings tangible relief and delivers measurable impact.**

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# Impact of the digital patient chart Meona

**– 29min**

**less Documentation per shift**

**+ 49%**

**higher System quality**

**+ 19%**

**improved Information quality**

**+ 3%**

**better Communication & Collaboration**

**+ 20%**

**higher User Satisfaction**

**- 9%**

**decreased Workload**

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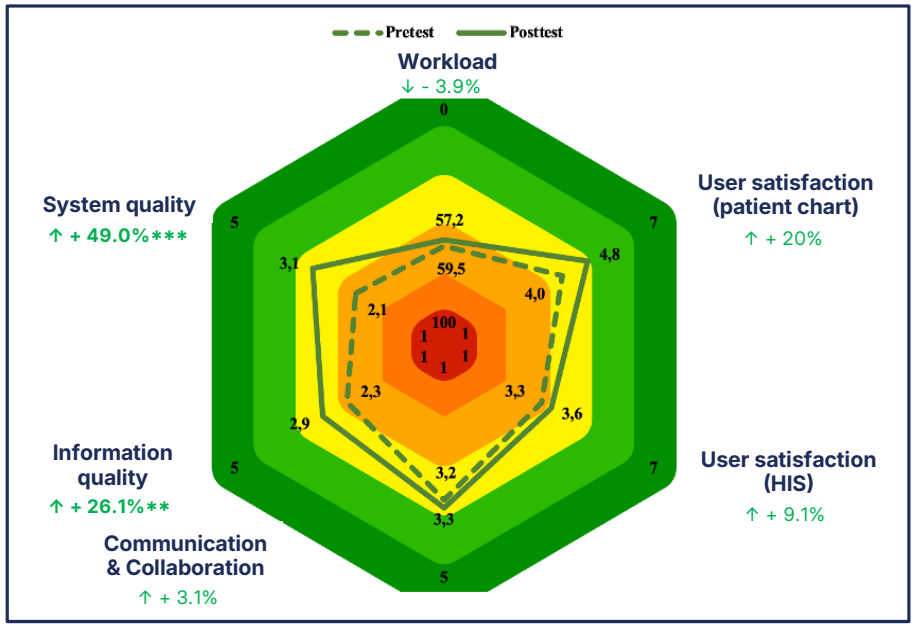
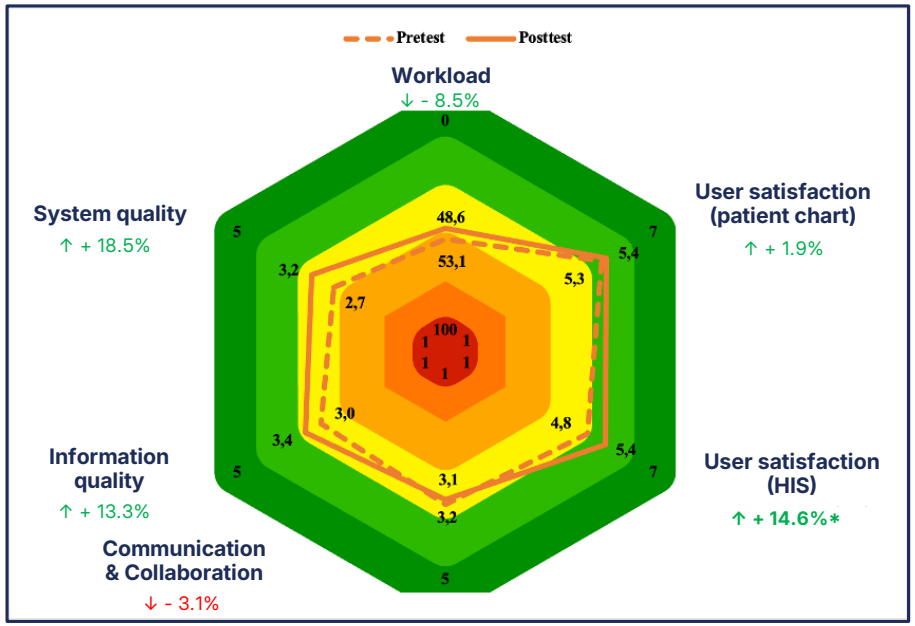
# Impact of the digital patient chart Meona

## Analog vs digital: Physicians[4]




## Analog vs digital: Nursing[4]




**Questionnaires with Likert scales from 1 ("poor") to 5 or 7 („good") and, for Workload, an inverted interval scale from 0 ("low") to 100 („high")**

*Based on the absolute changes: *p<0.05; **p<0.01; ***p≤$\alpha_{BH}$. $\alpha_{BH}$ denotes the Benjamini–Hochberg-corrected significance level (0.003).*

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## Impact of the digital patient chart Meona

**It’s not the systems that need our time**
**– patients do.**

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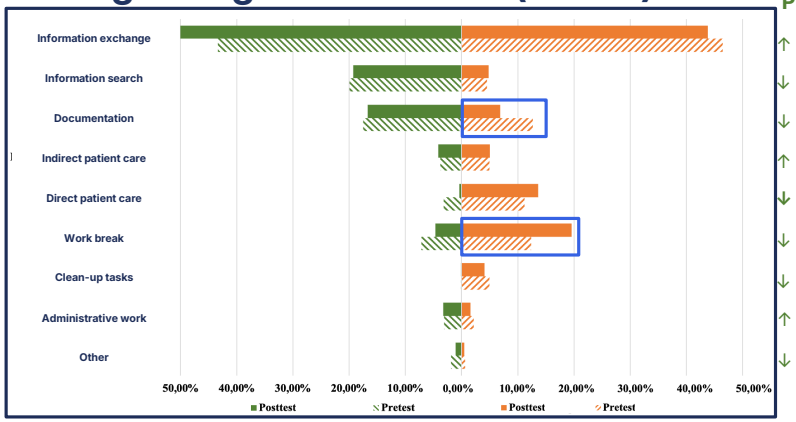
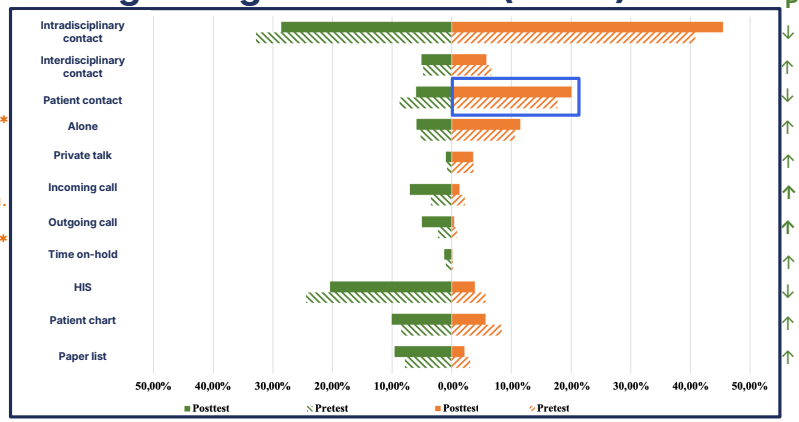
# Impact of the digital patient chart Meona

## Analog vs digital: activities (what?)[4]



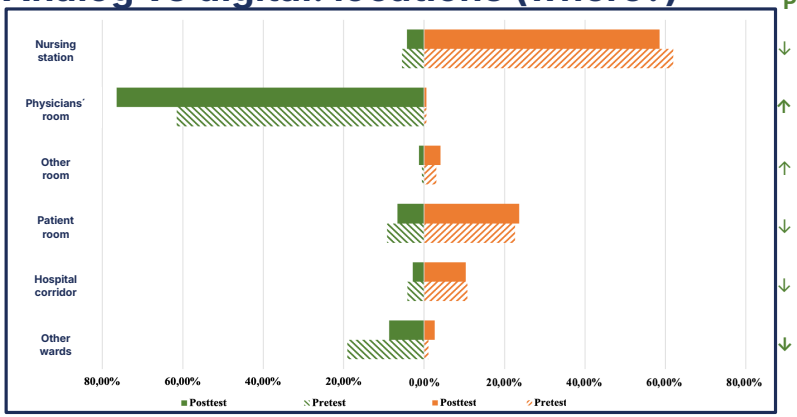

| Physicians | Nursing |
|---|---|
| ↑ + 39 min. | ↓ - 13 min. |
| ↓ - 3 min. | ↑ + 1 min. |
| ↓ - 4 min. | **↓ - 29 min.*** |
| ↑ + 2 min. | ↓↑ 0 min. |
| **↓ - 16 min.**** | ↑ + 12 min. |
| ↓ - 14 min. | **↑ + 36 min.**** |
| ↓ - 1 min. | ↓ - 4 min. |
| ↑ + 1 min. | ↓ - 3 min. |
| ↓ - 5 min. | ↓↑ 0 min. |

## Analog vs digital: medium (how?)[4]




| Physicians | Nursing |
|---|---|
| ↓ - 23 min. | ↑ + 23 min. |
| ↑ + 2 min. | ↓ - 4 min. |
| ↓ - 16 min. | ↑ + 12 min. |
| ↑ + 4 min. | ↑ + 4 min. |
| ↑ + 1 min. | ↓↑ 0 min. |
| **↑ + 19 min.*** | **↓ - 4 min.*** |
| **↑ + 15 min.*** | ↓ - 2 min. |
| ↑ + 2 min. | ↓↑ 0 min. |
| ↓ - 22 min. | ↓ - 5 min. |
| ↑ + 9 min. | **↓ - 13 min.*** |
| ↑ + 9 min. | **↓ - 5 min.*** |

## Analog vs digital: locations (where?)[4]




| Physicians | Nursing |
|---|---|
| ↓ - 7 min. | ↓ - 17 min. |
| **↑ + 83 min.**** | ↓↑ 0 min. |
| ↑ + 4 min. | ↑ + 5 min. |
| ↓ - 14 min. | ↑ + 5 min. |
| ↓ - 8 min. | ↓ - 2 min. |
| **↓ - 58 min.**** | ↑ + 8 min. |

**Nursing workday: 8:15 h (10% = 49.5 min.)**

**Physician workday: 9:15 h (10% = 55.5 min.)**

*Based on the absolute changes: *p<0.05; **p<0.01; ***p≤$\alpha_{BH}$.*
*$\alpha_{BH}$ denotes the Benjamini–Hochberg-corrected significance level (0.003).*

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# Successes and potentials of the digital patient chart Meona

**Information quality**

***Impact:*** **more relevant** information (physicians), **more readable/less error-prone** content (nursing)

***Potential:*** **paper list** remains the preferred information tool in both professional groups

**System quality**

***Impact:*** physician rating rises clearly – mainly due to **portability, user friendliness and effectiveness (eMAR)**

***Potential:*** bulky ward round charts and poor Wi-Fi coverage as **hardware barriers**

**User satisfaction**

***Impact:*** Digital patient chart is accepted – **no return** to paper

***Potential:*** **Usability hurdles** prevent noticeable day-to-day relief

**Communication & Collaboration**

***Impact:*** more **transparency** through digital entries, **instantaneous** delegation

***Potential:*** „**Illusion of Communication**" threatens SDM – digital entries do not replace real dialogue

**Workload**

***Impact:*** **no additional workload** from the new system

***Potential:*** **Subjective high time pressure remains**, especially in nursing

**Work processes**

***Impact:*** **–30 minutes documentation time** per shift → measurable relief for nursing through the digital chart

***Potential:*** Media breaks and **parallel paper lists** prevent real process integration

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**Good systems blend into hospital workflows.**
**Bad ones stand in your way.**

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# IMHIS®: Evidence that pays off

**Evidence for decisions:** **hard numbers** instead of gut feeling

- What actually reaches clinical staff from vendor promises?
- Example: −29 minutes nursing documentation per shift
- Assumption: €20/h, 6 wards, 10 nurses per ward
- **Projected annual savings: 11,000 hours = 5 FTE = €220,000**
- IMHIS® converts digital promises into **real impact** – contractually measurable KPIs

**Minimize risk and reveal process levers:** open the Black Box of daily hospital work

- Where impact is lost and where it's created – **and how to use it**
- **Identify** weaknesses, **build** on strengths, and **secure** the product's digital impact
- Clearly prioritize media breaks, usability and infrastructure – a **checklist for measurable success**

**Secure and communicate adoption:** vendor-neutral, evidence-based effectiveness from the user perspective

- User satisfaction as an **early-warning system** for daily hospital work
- Periodic quality reports and Governance – a **new perspective** beyond go-live metrics
- Priorities and product-insights from the **user perspective** – not from the whiteboard

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# IMHIS®: Evidence that relieves

**Evidence for daily hospital work:** digital systems that relieve instead of burden staff

- More time for what truly matters: **our patients**
- Information, when, where and how we need it – **when seconds count**
- Powerful, smart systems that run – **instead of clicking, waiting and swearing**

**Make digital impact visible:** open the Black Box of daily hospital work

- Where impact is lost and where it's created – **and how it can be maximized**
- **Identify** weaknesses, **build** on strengths, and **secure** the product's digital impact
- Media breaks, usability and infrastructure – what really frustrates hospital staff and what we need

**Check and communicate adoption:** transparently show what works and what doesn't

- User satisfaction as an **early-warning system** for daily hospital work
- Periodic quality reports and Governance – a **new perspective** beyond go-live metrics
- Priorities and product-insights from the **user perspective** – not just from the conference table

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# IMHIS®: Evidence that creates trust

**Marketing backed by real-world impact:** vendor-neutral, evidence-backed **effectiveness from the user perspective**

*Case study: digital patient chart:*

- **−30 minutes nursing documentation** per shift
- **+49% perceived system quality** among physicians
- Co-marketing with Dr. med. Florian Eisold: LinkedIn, use cases and target-group proof

**Product insights straight from users:** The Black Box of daily hospital work

- 99 standardized items plus 26 observation categories show how your system is perceived and used in daily work.
- Identify weaknesses, Build on strengths, and secure your product's digital impact

**Clear ROI case and story-telling for sales: what really lands with clinical staff?**

- Digital impact measurable across six dimensions: from system and information quality to user satisfaction, processes and workload
- Sharpen and win **tenders and reference stories**

**Priority roadmap:** What truly relieves. What slows you down.

- Concrete levers for your **product development** as well as the specific **Use-Cases** of your customers

**Trust asset: "Real Digital Impact":** quality seal and benchmarking for your product

*Draft*

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# IMHIS®: Make digital impact visible

**Excellence in care needs excellence in systems.**
**The digital future of care is proven**
**– not promised.**

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# IMHIS

## The digital future of healthcare starts with the right analysis

Let's measure together what truly works

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# Sources

# Sources

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